Antipodal Necessity and the Cubic Case of the Homogeneous Rotation-Symmetric Bent Function Conjecture

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Abstract

We prove a structural antipodal necessity theorem for low-degree rotation-symmetric bent functions. In every positive even dimension n, if a rotation-symmetric Boolean function f is bent and $\deg f \leq 3$, then its algebraic normal form contains the antipodal quadratic orbit

$$P_n(x) = \sum_{i=0}^{n/2-1} x_i x_{i+n/2}.$$

As an immediate consequence, no homogeneous rotation-symmetric Boolean function of algebraic degree three is bent in any positive even dimension, settling the cubic case of the homogeneous rotation-symmetric bent-function conjecture. The proof is algebraic and uniform in the dimension. Writing $n = 2^s m$ with m odd, we first transfer bentness to the fixed space of the odd-order rotation σ^{2^s}. In the resulting power-of-two dimension, a fiber argument forces the antipodal orbit. Half-turn cancellation transports the obstruction caused by a missing antipodal coefficient back to the original dimension. The universal theorem and its homogeneous cubic corollary have been kernel-checked in Lean 4.15.0. No computational classification is used in this proof.

Keywords: Boolean functions; bent functions; rotation symmetry; antipodal quadratic orbit; homogeneous Boolean functions; derivatives; fixed spaces.

1 Introduction and relation to earlier work

Rotation-symmetric Boolean functions form a natural class in symmetric cryptographic constructions: cyclic invariance gives a compact algebraic description while retaining nontrivial Walsh-spectral behavior. A long-standing conjecture of Stănică and Maitra [1] asserts that a homogeneous rotation-symmetric bent function cannot have algebraic degree greater than two. The conjecture has motivated a sequence of partial nonexistence results. Meng, Chen and Fu [2] obtained lower-degree restrictions; Zhang and Gao [3] derived further support-dependent exclusions; and Cusick and Sanger [4] proved substantial cases in dimensions $n = 2p$ together with an antipodal necessity result for quadratic rotation-symmetric bent functions and related short-cycle families.

Our main result is stronger than the homogeneous cubic nonexistence statement. For every positive even n, we prove

$$\boxed{f \text{ rotation-symmetric and bent}, \quad \deg f \leq 3 \quad \Longrightarrow \quad [P_n]f = 1,}$$

where

$$P_n(x) = \sum_{i=0}^{n/2-1} x_i x_{i+n/2}$$

is the unique antipodal quadratic orbit. A homogeneous cubic has no quadratic ANF terms, so the cubic case of the homogeneous conjecture follows immediately.

The proof separates into three structural steps. First, write $n = 2^s m$ with m odd and restrict to the fixed space of the odd-order rotation σ^{2^s}. For functions of degree at most three, bentness survives this restriction because every nonzero derivative is a balanced quadratic and balance of an invariant quadratic transfers exactly to the odd-order fixed space. Second, in power-of-two dimension we prove an Antipodal Rule: every rotation-symmetric bent function of degree at most three must contain the antipodal quadratic orbit. Its proof combines a diagonal/fiber decomposition, Parseval, rigidity of balanced rotation-symmetric quadratics, third differences, and cyclic covariance of a complementary fiber. Third, if the original antipodal coefficient were zero, half-turn cancellation would make the original function constant on its half-turn fixed space. Repeating a diagonal vector from the smaller space lies in this fixed space, so the restricted bent function would have constant diagonal, contradicting the power-of-two fiber argument. The stronger coefficient-transport identity in Section 4 is retained as an auxiliary algebraic result; it is neither needed by this proof route nor included in the kernel certification. These three ingredients yield the universal antipodal theorem and then the homogeneous cubic corollary.

The closest recent overlap is Sun, Shi, Liu and Fu [5]. Their 2026 paper explicitly treats cubic homogeneous rotation-symmetric functions, but its cubic theorems remain conditional: their Theorem 5 assumes additional parity and 2-adic conditions on the generators in the short algebraic normal form (SANF), their Theorem 6 excludes divisibility subfamilies of dimensions for each fixed generator pattern, and their Corollary 2 imposes strong arithmetic restrictions on the odd prime factors of n. Their conclusion states that the general conjecture remains open. Our argument removes these arithmetic and support restrictions in degree three. Within the literature reviewed for this manuscript, we have not identified an earlier theorem excluding homogeneous cubic rotation-symmetric bent functions uniformly in every positive even dimension, nor an earlier antipodal necessity theorem for arbitrary rotation-symmetric bent functions of algebraic degree at most three. This is a scope statement rather than an absolute priority claim.

Another 2026 development is the work of Polujan, Kudin and Pašalić [8], who classify rotation-symmetric cubic bent functions in ten variables under extended-affine equivalence and exhibit a class outside the completed Maiorana–McFarland class. Their result concerns general, nonhomogeneous cubic rotation-symmetric bent functions and is complementary to the homogeneous nonexistence problem treated here.

Nonhomogeneous cubic rotation-symmetric bent functions are known. Gao, Zhang, Liu and Carlet [7] construct an infinite family in the Maiorana–McFarland class whose ANF contains the antipodal quadratic coupling, and Wang et al. [6] construct power-of-two-variable cubic rotation-symmetric bent families with the same structural feature. These positive constructions are consistent with the theorem proved here: cubic rotation symmetry and bentness can coexist, but in degree at most three the antipodal quadratic component is unavoidable.

2 Notation

Let $V_n = \mathbb{F}_2^n$. The algebraic normal form (ANF) of a Boolean function is its unique multilinear polynomial over $\mathbb{F}_2$. Homogeneity of degree three means that every nonzero ANF term has degree three. The zero function may be included in spans without affecting any nonexistence assertion. For $a \in V_n$, set $D_a f(x) = f(x+a) + f(x)$. Define

$$W_f(b) = \sum_{x \in V_n} (-1)^{f(x)+b\cdot x}, \qquad W_f(0) = 2^n - 2\,\mathrm{wt}(f).$$

A function is balanced if $W_f(0) = 0$, and bent if $|W_f(b)| = 2^{n/2}$ for every b. Equivalently, every nonzero directional derivative is balanced [10]. Indeed, the Fourier transform of the autocorrelation $A_f(a) = \sum_x (-1)^{D_a f(x)}$ is $W_f(b)^2$. Thus constant squared Walsh magnitude is equivalent to $A_f(0) = 2^n$ and $A_f(a) = 0$ for $a \neq 0$.

Let σ be cyclic rotation of the coordinates. Rotation symmetry means $f \circ \sigma = f$. Such functions are linear combinations of sums over distinct cyclic monomial orbits. All polynomial sums are in $\mathbb{F}_2$; exponential sums and weights are ordinary integers.

For a Boolean function q, define its normalized polar expression by

$$B_q(u, v) = q(u + v) + q(u) + q(v) + q(0).$$

When $\deg q \leq 2$, the map B_q is alternating bilinear and is unchanged by adding a constant to q.

For even n, define the antipodal quadratic orbit

$$P_n(x) = \sum_{i=0}^{n/2-1} x_i x_{i+n/2}.$$

If f is rotation-symmetric, its ANF is a sum of distinct cyclic orbit sums. We write $[P_n]f \in \mathbb{F}_2$ for the coefficient of the orbit sum P_n in this decomposition.

3 Odd-order fixed-space reduction

Lemma 3.1. *Let V be a finite-dimensional vector space over $\mathbb{F}_2$, and let $T \in \mathrm{GL}(V)$ satisfy $T^m = I$ for an odd positive integer m. If $q : V \to \mathbb{F}_2$ has degree at most two and $q \circ T = q$, then q is balanced on V if and only if $q|_U$ is balanced on $U = \mathrm{Fix}(T)$.*

Proof. Adding $q(0)$ preserves balance both on V and on U, so assume $q(0) = 0$. Write $B = B_q$. By the definition above,

$$B(x, y) = q(x + y) + q(x) + q(y)$$

under this normalization; it is alternating bilinear and T-invariant. Set $P = \sum_{j=0}^{m-1} T^j$. Then $TP = PT = P$, $Pu = u$ for $u \in U$, and $P^2 = P$. Hence $V = U \oplus K$, where $K = \ker P$. For $u \in U$ and $k \in K$, T-invariance gives $B(u, T^j k) = B(T^{-j}u, k) = B(u, k)$. Since m is odd,

$$B(u, k) = mB(u, k) = \sum_{j=0}^{m-1} B(u, T^j k) = B(u, Pk) = 0.$$

It follows that $q(u + k) = q(u) + q(k)$, and thus

$$S_V(q) = S_U(q|_U)S_K(q|_K), \qquad S_E(h) = \sum_{x \in E} (-1)^{h(x)}.$$

We show $S_K(q|_K) \neq 0$. Put $R = \{z \in K : B(z, k) = 0 \text{ for all } k \in K\}$. Since P commutes with T, K is invariant under T and T^{-1}. If $z \in R$ and $k \in K$, then $B(Tz, k) = B(z, T^{-1}k) = 0$, so R is T-invariant. On R, the function q is linear. More explicitly, if $z \in R$, then all $T^j z$ lie in R, so every cross-polar term $B(T^i z, T^j z)$ vanishes. Hence

$$q(Pz) = q\left(\sum_{j=0}^{m-1} T^j z\right) = \sum_{j=0}^{m-1} q(T^j z) = mq(z) = q(z).$$

Since $z \in K$, $Pz = 0$, and therefore $q(z) = q(0) = 0$. Thus the only possible obstruction to a nonzero quadratic character sum on K disappears: q vanishes identically on the radical of its polar form. Character orthogonality now gives

$$\begin{aligned} S_K(q|_K)^2 &= \sum_{z \in K} (-1)^{q(z)} \sum_{x \in K} (-1)^{B(x,z)} \\ &= |K| \sum_{z \in R} (-1)^{q(z)} = |K||R| > 0. \end{aligned}$$

The factorization proves the equivalence of balance. □

Proposition 3.2. *Under the hypotheses on T above, let $f \circ T = f$, $\deg f \le 3$, and suppose f is bent. Then every nonzero derivative of $p = f|_U$ is balanced. Consequently $\dim U$ is even and p is bent (with the usual spectral convention also covering $\dim U = 0$).*

Proof. For each $a \in U \setminus \{0\}$, $D_a f$ is quadratic, balanced, and T-invariant: $Ta = a$ implies $D_a f(Tx) = D_a f(x)$. Lemma 3.1 shows that $(D_a f)|_U = D_a(f|_U)$ is balanced. If $d = \dim U$ and

$$C_p(a) = \sum_{x \in U} (-1)^{p(x)+p(x+a)},$$

then $C_p(0) = 2^d$ and $C_p(a) = 0$ for $a \neq 0$. Fourier inversion of autocorrelation gives

$$W_p(b)^2 = \sum_{a \in U} C_p(a)(-1)^{b \cdot a} = 2^d.$$

Thus d must be even (unless $d = 0$), because Walsh coefficients are integers, and $|W_p(b)| = 2^{d/2}$ for every b. Hence p is bent. □

4 Exact antipodal coefficient transport under folding

Scope. This section is an auxiliary algebraic argument retained from the earlier manuscript. Its full coefficient identity is not Lean-formalized and is not used in the certified proof of the main theorem below.

Write $n = tm$, where $t = 2^s$, $s \ge 1$, and m is odd. The map $T = \sigma^t$ has order m, and its fixed subspace consists of repetitions of a block $y \in \mathbb{F}_2^t$. Under the identification $\mathrm{Fix}(T) \cong \mathbb{F}_2^t$, the restriction is the folding substitution

$$x_i = y_{i \bmod t}.$$

Moreover, $\mathrm{Fix}(T)$ is σ-stable, and the action induced by σ on the block coordinates is the cyclic shift on $\mathbb{F}_2^t$. Hence the restriction of a rotation-symmetric function is again rotation-symmetric. The following lemma is the precise orbit-counting statement needed later.

Lemma 4.1 (Exact antipodal coefficient transport)**.** *Let $n = tm$, where $t = 2^s$ with $s \ge 1$ and m is odd. Let $f : \mathbb{F}_2^n \to \mathbb{F}_2$ be rotation-symmetric with $\deg f \le 3$, and let g be its restriction under $x_i = y_{i \bmod t}$. Then*

$$\boxed{[P_t]g = [P_n]f.}$$

Proof. By linearity it suffices to fold one distinct cyclic monomial orbit. Let $A \subseteq \mathbb{Z}_n$ be a nonempty support with $|A| \le 3$, let

$$h = |\operatorname{Stab}_{\mathbb{Z}_n}(A)|, \qquad B = \{a \bmod t : a \in A\}, \qquad b = |\operatorname{Stab}_{\mathbb{Z}_t}(B)|.$$

Write O_A, O_B for the sums over distinct translates.

Reduction modulo t is equivariant for the natural homomorphism

$$\mathbb{Z}_n \longrightarrow \mathbb{Z}_t.$$

Hence it induces a surjective map from the distinct monomials in O_A onto those in O_B. Since both source and target are transitive cyclic orbits, all fibers have the same cardinality. Therefore every monomial in O_B occurs exactly

$$\frac{|O_A|}{|O_B|} = \frac{n/h}{t/b} = \frac{mb}{h}$$

times. In particular, this ratio is automatically an integer, even when collisions lower the degree of the reduced monomial. Thus uniformly for every nonempty A with $|A| \le 3$,

$$\boxed{\mathrm{Fold}(O_A) = \left(\frac{mb}{h} \bmod 2\right) O_B.} \tag{$*$}$$

It remains only to identify when the target orbit O_B can be P_t. The stabilizer of A acts freely on A, so $h \mid |A|$: in degree one $h = 1$; in degree two $h = 1$ except for the antipodal orbit P_n, where $h = 2$; and in degree three $h = 1$ or 3.

If a non-antipodal quadratic orbit reduces to an antipodal pair, then $h = 1$ and $b = 2$, so the multiplicity in $(*)$ is $2m$, hence even. If a cubic orbit reduces, after collisions, to an antipodal pair, then again its target has $b = 2$. Such an orbit cannot have $h = 3$: the case $h = 3$ is the short cubic support

$$\{a, a + n/3, a + 2n/3\},$$

which requires $3 \mid m$ and collapses modulo t to a singleton. Hence every cubic orbit that reduces to an antipodal pair has $h = 1$, so its multiplicity is again $2m$ and it cancels.

Finally consider P_n. Here $h = 2$. Since

$$\frac{n}{2} = \frac{tm}{2} \equiv \frac{t}{2} \pmod{t}$$

because m is odd, its reduced support is antipodal, so $b = 2$. Formula $(*)$ gives multiplicity

$$\frac{mb}{h} = \frac{2m}{2} = m,$$

which is odd. Therefore

$$\mathrm{Fold}(P_n) = P_t.$$

Constants and linear terms cannot contribute a quadratic term. Hence P_n is the unique original orbit whose fold contributes nontrivially to P_t, and consequently

$$[P_t]g = [P_n]f.$$

□

5 Antipodal necessity in power-of-two dimension

We first isolate the quadratic input, then compare the diagonal and complementary fibers of a low-degree rotation-symmetric bent function.

Lemma 5.1 (Quadratic rotation-symmetric rigidity)**.** *Let $N = 2^k$. If $q : \mathbb{F}_2^N \to \mathbb{F}_2$ is rotation-symmetric, balanced, and $\deg q \le 2$, then the polar form of q is zero.*

Proof. Adding a constant does not affect balance or the polar, so put $Q = q + q(0)$, with $Q(0) = 0$, and let B be its polar. Let $R = \{r : B(r, x) = 0 \ \forall x\}$. If $B \neq 0$, then R is a proper rotation-invariant subspace.

Identify $\mathbb{F}_2^N$ with $A = \mathbb{F}_2[X]/((X+1)^N)$, with cyclic rotation S represented by multiplication by X. Since A is local with maximal ideal $(X+1)$, a vector of odd Hamming weight represents a unit: its polynomial evaluates to 1 at $X = 1$. If a rotation-invariant subspace contains a unit u, it contains $u, Xu, \ldots, X^{N-1}u$. Multiplication by u is an automorphism of A, while $1, X, \ldots, X^{N-1}$ span A; consequently $u, Xu, \ldots, X^{N-1}u$ also span A. Thus every proper rotation-invariant subspace contains no unit, equivalently no odd-weight vector, and

$$R \subseteq (X + 1)A = \mathrm{im}(S + I).$$

For $r \in R$, write $r = Sy + y$. Rotation invariance gives $Q(Sy) = Q(y)$, so

$$Q(r) = Q(Sy) + Q(y) + B(Sy, y) = B(r + y, y) = 0,$$

because $B(r, y) = B(y, y) = 0$.

Finally,

$$\left(\sum_x (-1)^{Q(x)}\right)^2 = \sum_z (-1)^{Q(z)} \sum_x (-1)^{B(x,z)} = 2^N \sum_{z \in R} (-1)^{Q(z)} = 2^N |R| > 0.$$

Thus Q, and hence q, is not balanced, a contradiction. Therefore $B = 0$. □

Remark 5.2 (Sharpness of the power-of-two hypothesis). Among even dimensions, the hypothesis $N = 2^k$ in Lemma 5.1 is exact. This sharpness is consistent with, and can also be deduced from, the known balancing theory for quadratic monomial rotation-symmetric functions; in particular, Chirvasitu and Cusick characterize when $(0, t)_N$ is balanced in terms of the exact power of two dividing N and t. We include the following direct radical argument because it matches the mechanism used here and makes no novelty claim for the balancing construction. If

$$N = dm, \qquad d = 2^{v_2(N)}, \qquad m > 1 \text{ odd},$$

then

$$q_d(x) = \sum_{i=0}^{N-1} x_i x_{i+d}$$

is rotation-symmetric and quadratic. Its underlying graph is the disjoint union of d odd cycles of length m. The radical of its polar consists of vectors constant independently on these cycles. Hence it has dimension d, while on a vector supported as 1 on one cycle,

$$q_d(r) = m = 1 \pmod 2.$$

Thus q_d restricts to a nonzero linear form on the radical. Using

$$\left(\sum_x (-1)^{q_d(x)}\right)^2 = 2^N \sum_{r \in \mathrm{rad}(B_{q_d})} (-1)^{q_d(r)},$$

the right-hand side vanishes, so q_d is balanced. Its polar is nonzero. Therefore, for even N, every balanced rotation-symmetric quadratic having zero polar is a rigidity phenomenon valid for all such functions precisely when N is a power of two.

Lemma 5.3 (Complementary-fiber balance). *Let $N = 2h$, let $g : \mathbb{F}_2^N \to \mathbb{F}_2$ be bent, and suppose $g(u, u)$ is constant. Define $F_z(u) = g(u, u + z)$. Then F_z is balanced for every $z \neq 0$.*

Proof. After adding a constant assume $g(u, u) = 0$. Put $A_z = \sum_u (-1)^{F_z(u)}$. Under the invertible linear change $\Phi(u, z) = (u, u + z)$, put $G(u, z) = g(\Phi(u, z)) = g(u, u + z)$. Then

$$\widehat{A}(b) = \sum_z A_z (-1)^{b \cdot z} = \sum_{u,z} (-1)^{G(u,z) + b \cdot z} = W_G(0, b).$$

Bentness is preserved by the invertible linear change Φ, so, since G has $N = 2h$ variables,

$$|\widehat{A}(b)| = |W_G(0, b)| = 2^{N/2} = 2^h$$

for every b. Parseval for the unnormalized Walsh transform on $\mathbb{F}_2^h$ therefore gives

$$\sum_z A_z^2 = 2^{-h} \sum_b \widehat{A}(b)^2 = 2^{-h} \cdot 2^h \cdot 2^{2h} = 2^{2h} = 2^N.$$

But $A_0 = 2^h$, so $A_0^2 = 2^N$ exhausts the sum. Hence $A_z = 0$ for every $z \neq 0$. □

Lemma 5.4 (Local complementary-fiber rigidity)**.** *Let $N = 2h$ be any positive even integer, and let $g : \mathbb{F}_2^N \to \mathbb{F}_2$ be rotation-symmetric with $\deg g \le 3$. Suppose $g(u, u)$ is constant. If the polar form of $D_J g$, where $J = \mathbf{1}^N$, is zero, then $F_{\mathbf{1}}(u) = g(u, u + \mathbf{1})$ is constant.*

Proof. Adding a constant, assume $g(u, u) = 0$. Put $d(u) = (u, u)$, $e(z) = (0, z)$, and $J = \mathbf{1}^N$. By hypothesis, $B_{D_J g} = 0$.

First, $F_{\mathbf{1}}$ has degree at most two. Indeed, translating the second block by the constant vector $\mathbf{1}$ changes a cubic monomial only by terms of degree at most two. Therefore the homogeneous cubic part of

$$u \longmapsto g(u, u + \mathbf{1})$$

is the same as the homogeneous cubic part of $u \mapsto g(u, u)$. Since $g(u, u)$ is constant, that cubic part vanishes.

For degree at most three,

$$\Delta_3(p, q, r) = D_p D_q D_r g(0)$$

is independent of the base point and is a symmetric trilinear form. Put $p = d(u)$, $q = d(v)$, $w = e(\mathbf{1})$. Then

$$B_{F_{\mathbf{1}}}(u, v) = D_p D_q g(w) = D_p D_q g(0) + \Delta_3(p, q, w).$$

Here

$$D_p D_q g(0) = g(0) + g(p) + g(q) + g(p + q) = 0,$$

because $0, p, q, p+q$ are all diagonal points and the normalized g vanishes on the diagonal. Thus

$$B_{F_{\mathbf{1}}}(u, v) = \Delta_3(d(u), d(v), e(\mathbf{1})).$$

Also

$$B_{D_J g}(d(u), e(v)) = \Delta_3(d(u), e(v), J).$$

Writing $e'(z) = (z, 0)$, half-rotation fixes $d(u)$, exchanges $e(v)$ with $e'(v)$, and exchanges $e(\mathbf{1})$ with $e'(\mathbf{1})$. Since $J = e'(\mathbf{1}) + e(\mathbf{1})$, trilinearity gives

$$\Delta_3(d(u), e(v), J) = \Delta_3(d(u), e(v), e'(\mathbf{1})) + \Delta_3(d(u), e(v), e(\mathbf{1})).$$

Applying the half-rotation simultaneously to all three arguments in the first term yields

$$\Delta_3(d(u), e(v), e'(\mathbf{1})) = \Delta_3(d(u), e'(v), e(\mathbf{1})).$$

A final use of trilinearity, with $d(v) = e(v) + e'(v)$, therefore gives

$$\Delta_3(d(u), e(v), J) = \Delta_3(d(u), d(v), e(\mathbf{1})).$$

Since $B_{D_J g} = 0$, the polar of $F_{\mathbf{1}}$ is zero; hence $F_{\mathbf{1}}$ is affine.

With $(\sigma x)_i = x_{i-1 \pmod N}$, direct calculation gives

$$\sigma(u, u + \mathbf{1}) = (Ru + e_0, Ru + e_0 + \mathbf{1}),$$

where R is cyclic rotation on $\mathbb{F}_2^{N/2}$. Thus $F_{\mathbf{1}}(Ru + e_0) = F_{\mathbf{1}}(u)$. If $F_{\mathbf{1}}(u) = a \cdot u + c$, then $R^T a = a$, so a is constant on the single cycle of R; the translation condition gives $a \cdot e_0 = 0$, forcing $a = 0$. Therefore $F_{\mathbf{1}}$ is constant. □

Theorem 5.5 (Antipodal Rule)**.** *Let $N = 2^k$ with $k \ge 2$. If $g : \mathbb{F}_2^N \to \mathbb{F}_2$ is rotation-symmetric, bent, and $\deg g \le 3$, then its ANF contains*

$$P_N(x) = \sum_{i=0}^{N/2-1} x_i x_{i+N/2}.$$

Proof. Write $N = 2h$, and let α be the coefficient of P_N. On the diagonal $d(u) = (u, u)$, linear orbit sums cancel in pairs, non-antipodal quadratic orbit sums cancel under half-rotation, and cubic orbit sums cancel because every cubic orbit has full length when N is a power of two. The antipodal orbit restricts to $\sum_i u_i$. Hence

$$g(u, u) = c + \alpha \sum_i u_i.$$

If $\alpha = 0$, the diagonal is constant. Lemma 5.3 makes $F_{\mathbf{1}}$ balanced. Moreover, bentness makes $D_J g$ balanced. Since $J = \mathbf{1}^N$ is fixed by cyclic rotation and g is rotation-symmetric, $D_J g$ is rotation-symmetric; also $\deg D_J g \leq 2$. Since $N = 2^k$, Lemma 5.1 gives $B_{D_J g} = 0$, and Lemma 5.4 then makes $F_{\mathbf{1}}$ constant. This is a contradiction. Therefore $\alpha = 1$. □

6 Universal antipodal necessity and the cubic homogeneous case

Theorem 6.1 (Universal antipodal necessity through degree three)**.** *Let n be a positive even integer. If $f : \mathbb{F}_2^n \to \mathbb{F}_2$ is rotation-symmetric, bent, and* $\deg f \leq 3$*, then*

$$[P_n]f = 1.$$

Equivalently, the ANF of f contains the antipodal quadratic orbit P_n.

Proof. Write $n = tm$, where $t = 2^s$, $s \geq 1$, and m is positive and odd. By Proposition 3.2, the function

$$g(y) = f(y, y, \ldots, y), \qquad y \in \mathbb{F}_2^t,$$

with m repeated blocks, is bent, rotation-symmetric, and of degree at most three.

Suppose $[P_n]f = 0$. The half-turn $i \mapsto i + n/2$ is a fixed-point-free involution on coordinate indices. Pair each ANF support with its half-turn image. Rotation symmetry gives equal coefficients, and the paired monomials agree on every half-turn fixed vector, so distinct pairs cancel. Among supports of size at most three, the only nonempty supports fixed by the half-turn are the antipodal pairs: an invariant support is a union of two-element coordinate orbits. Their coefficients vanish by the supposition and rotation symmetry. Consequently $f(x) = f(0)$ on the entire half-turn fixed space.

For $y = (u, u) \in \mathbb{F}_2^t$, the repeated vector has period $t/2$. Since $n/2 = m(t/2)$, it is fixed by the original half-turn. Thus $g(u, u) = g(0)$ for every u. If $t \geq 4$, apply Lemma 5.3 to g. The nonzero all-one derivative is balanced by bentness, rotation-symmetric, and quadratic. Lemma 5.1 makes its polar zero, and Lemma 5.4 makes the complementary fiber constant, contradicting its balance.

If $t = 2$, write the rotation-symmetric ANF as $g(y_0, y_1) = c + a(y_0 + y_1) + by_0y_1$. Constancy on (u, u) forces $b = 0$, making g affine and hence not bent. This also contradicts bentness. Therefore $[P_n]f = 1$ in every positive even dimension. □

Theorem 6.2 (Main theorem)**.** *For every positive even integer n, there is no homogeneous rotation-symmetric bent Boolean function of degree three in n variables.*

Proof. If such a function f existed, Theorem 6.1 would force $[P_n]f = 1$. Homogeneity of degree three forces every quadratic ANF coefficient, in particular $[P_n]f$, to be zero. This contradiction proves the theorem. □

7 Discussion

The universal antipodal theorem isolates the structural reason that homogeneous cubics fail. Homogeneity is not used in the power-of-two Antipodal Rule or in the odd-order fixed-space

transfer; it enters only at the final step, where a homogeneous cubic cannot contain the quadratic orbit that every low-degree rotation-symmetric bent function is forced to contain.

The proof also identifies the point at which the present method is genuinely degree-sensitive. For $\deg f \leq 3$, every derivative $D_a f$ is quadratic, so the odd-order fixed-space balance-transfer lemma applies. For degree at least four, derivatives need not be quadratic, and no analogous fixed-space transfer theorem is currently available from this argument alone. Thus the cubic theorem does not by itself extend to higher degree.

The quadratic rigidity input is compatible with the established theory of quadratic rotation-symmetric Boolean functions developed by Chirvasitu and Cusick [9]; we make no separate novelty claim for the known balancing phenomena underlying that auxiliary step. The new contribution is the combination of fixed-space transfer, power-of-two antipodal rigidity, and half-turn obstruction transport, culminating in the all-even-dimensional degree-at-most-three antipodal necessity theorem.

8 Verification scope and reproducibility

The universal antipodal theorem and homogeneous cubic nonexistence corollary are now kernel-checked in Lean 4.15.0, with Mathlib pinned to commit `9837ca9d`. The final proof source is commit `6dd18368`; the accompanying reports are at commit `6f9f66c2` of hrsbf-bent-verification. The latter changes documentation only. The 60-file integrity manifest was checked without discrepancies when preparing this revision.

The formal definitions use $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$, vectors indexed by $\mathrm{Fin}(n)$, actual cyclic rotation, and canonical squarefree ANF coefficients obtained by Boolean Möbius inversion. The degree hypothesis says that every support of size greater than three has coefficient zero. Constants and affine terms are unrestricted. For every $H > 0$, the theorem on $2H$ variables assumes exactly rotation symmetry, degree at most three, and

$$\left| \sum_{x \in \mathbb{F}_2^{2H}} (-1)^{f(x)+b\cdot x} \right| = 2^H \quad \text{for every } b,$$

and concludes that the coefficient of $x_0 x_H$ is one. Rotation symmetry identifies this with the coefficient of P_{2H}. The homogeneous corollary requires all noncubic coefficients to vanish and at least one cubic coefficient to be nonzero. The zero function is separately non-bent in positive dimension. The opposite cyclic orientation used in some coordinate formulas generates the same cyclic group and hence the same invariance condition.

The public Lean names, in namespace `HRSBF.Antipodal`, are

```
universal_antipodal_walsh_absolute
no_homogeneous_cubic_rotation_symmetric_bent_absolute.
```

Their axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`; neither `sorryAx` nor native-evaluation axioms occur. The real GitHub Actions logs for run 38682481132 were retrieved and inspected during this publication preparation: the complete library build, explicit theorem signatures, and axiom checks passed at the exact source commit above. The earlier core proof also passed in run 38682075049. This revision does not claim a new local Lean execution. The proof uses odd-order fixed-space restriction and half-turn obstruction transport, matching the proof of Theorem 6.1 above. The full folding-coefficient identity of Section 4, the literature comparisons, and the computational reports below are outside that kernel certification. The archived research document named `NOVO_TEOREMA_V2` is a separate, unresolved research project and is not incorporated into this revision.

For reproduction, use the pinned source commit and enter `formalization`. Obtain the pinned Mathlib cache with `lake exe cache get`. Then run `lake build` and `lake env lean`

`CheckGlobal.lean`. The accompanying supplement preserves the source, workflow, dependency pins, signatures, logs, and integrity manifest. Kernel verification is not external peer review or a certificate of bibliographic priority.

The following paragraph describes the earlier, separate finite-example formalization, retained as historical supporting material rather than as the proof of the universal theorem.

The archived self-contained Lean 4.15.0 file certifies an explicit nonhomogeneous rotation-symmetric cubic bent function in twelve variables, including its actual Walsh definition, ANF degree and nonhomogeneity, and weight 2080. It also certifies the transfer-matrix trace $\mathrm{Tr}(M^{32}) = 85\,032\,960$ and several general algebraic and finite-sum lemmas. The source compiles with exit code zero and its 543 declared theorems have axiom reports. Its axiom reports contain only the standard logical axioms of Lean: propositional extensionality, quotient soundness, and classical choice. There are no admitted proof holes or axioms admitting native evaluation. The bit-walk identification for H_{32} remains an explicit hypothesis in that formal file and is not used as a premise of the global algebraic proof above.

Independent exact controls exhaust all 2^{16} quadratic rotation-symmetric components in dimension 32 and both choices of the invariant linear component. None of the resulting 131 072 normalized functions is balanced with nonzero polar. Both constant terms follow immediately, since adding a constant preserves balance. For the 155 cubic orbits, the controls verify the third-difference contraction on all 16×16 basis pairs, and equality of the row spaces of the complementary-fiber polar and nonconstant coefficient maps. Both have rank six; their common kernel has dimension 149.

In dimension $n = 12$, an exhaustive CUDA search on an NVIDIA A100 GPU evaluated all $2^{25} = 33\,554\,432$ rotation-symmetric functions of algebraic degree at most three (generated by 6 quadratic and 19 cubic monomial orbits, modulo invariant affine terms). Exactly 14 336 affine representatives (57 344 functions including constant and linear terms) are bent. Every single bent function contains the antipodal quadratic orbit P_{12}, and no homogeneous cubic rotation-symmetric bent function exists in twelve variables, confirming zero violations of the antipodal necessity theorem. In dimension $n = 14$, an extensive search on the same NVIDIA A100 GPU evaluated 500 006 912 rotation-symmetric functions of degree at most three (> 500 million candidates), a deterministic prefix covering only 5.82% of the 2^{33} coefficient combinations; the run is incomplete. Among 25 176 bent representatives identified (100 704 with affine terms), exactly 100% contain the antipodal quadratic orbit P_{14}, with zero bent functions lacking P_{14} and zero homogeneous cubic bent functions, recording no violations within that tested prefix. These finite checks are consistency controls, not a replacement for the uniform proof.

The twelve-variable example is a positive control illustrating why homogeneity matters. It does not contradict the homogeneous hypotheses in the cited nonexistence results, and no novelty claim is made for its membership in the Maiorana–McFarland class. This version is a public preprint for expert scrutiny, not an accepted journal article and not an assertion of independent external peer review.

Declaration of AI-assisted research and manuscript preparation

Generative AI systems assisted proof exploration, adversarial checking, verification-code development, literature searches, and manuscript editing. Earlier versions used OpenAI ChatGPT, Google Gemini, and Anthropic Claude. This revision was consolidated with OpenAI Codex, with formal claims checked against the stated Lean source and actual CI logs. AI output is not itself mathematical evidence; the reproducible proof artifacts delimit the kernel-verified claims. AI systems are not authors. Responsibility for the published manuscript remains with the named author.

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